

ID: 14661

NAME: RIDA BILAL

DEPARTMENT: BS(SE)

SEMESTER: 5th

ASSIGNMENT

1.

```
ubuntu@ubuntu:~$ cd Desktop
ubuntu@ubuntu:~/Desktop$ cd c++_programs
bash: cd: c++_programs: No such file or directory
ubuntu@ubuntu:~/Desktop$ ps
  PID TTY          TIME CMD
  5365 pts/0    00:00:00 bash
  5377 pts/0    00:00:00 ps
ubuntu@ubuntu:~/Desktop$ ps -A
ps -A: command not found
ubuntu@ubuntu:~/Desktop$ ps -A
  PID TTY          TIME CMD
    1 ?            00:00:04 systemd
    2 ?            00:00:00 kthreadd
    3 ?            00:00:00 pool_workqueue_release
    4 ?            00:00:00 kworker/R-rcu_gp
    5 ?            00:00:00 kworker/R-sync_wq
    6 ?            00:00:00 kworker/R-kvfree_rcu_reclaim
    7 ?            00:00:00 kworker/R-slub_flushwo
```

2.

```
5328 ?            00:00:00 kworker/u4:0-events_power_efficient
5353 ?            00:00:00 gnome-terminal-
5365 pts/0    00:00:00 bash
5388 pts/0    00:00:00 ps
ubuntu@ubuntu:~/Desktop$ ps -lx
 F  UID      PID     PPID  PRI  NI     VSZ   RSS WCHAN  STAT TTY          TIME COMMAND
 4  1000    2168         1   20   0   21736 13228 -        Ss   ?            0:01 /usr/l
 5  1000    2169    2168   20   0   21460  3616 -        S    ?            0:00 (sd-pa
 4  1000    2182    2168   20   0 124200 13780 ep_pol Ssl  ?            0:00 /usr/bi
 4  1000    2183    2168   20   0 106804  5760 ep_pol Ssl  ?            0:00 /usr/bi
 4  1000    2185    2168   20   0 415908 16432 do_pol Ssl  ?            0:00 /usr/bi
 4  1000    2186    2168   20   0 124544 11536 ep_pol Ssl  ?            0:00 /usr/bi
 4  1000    2187    2168   20   0 325580  8600 do_pol Ssl  ?            0:00 /usr/bi
 4  1000    2192    2168   20   0  10692  6408 ep_pol Ss   ?            0:00 /usr/bi
 4  1000    2211    2158   20   0 244808  6132 do_pol Ssl+ tty2         0:00 /usr/l
 4  1000    2213    2211   20   0 315524 90500 ep_pol Sl+  tty2         0:13 /usr/l
 4  1000    2238    2211   20   0 307332 14004 do_pol Sl+  tty2         0:00 /usr/l
 4  1000    2320    2168   20   0 382992  8388 do_pol Ssl  ?            0:00 /usr/l
 4  1000    2327    2320   20   0   9480  5080 ep_pol S    ?            0:00 /usr/bi
 4  1000    2337    2168   20   0 162652  7116 do_pol Ssl  ?            0:00 /usr/l
 4  1000    2338    2168   20   0 100644  5596 do_pol Ssl  ?            0:00 /usr/l
 4  1000    2345    2168   20   0 333300  8800 do_pol Ssl  ?            0:00 /usr/l
```

3.

```
0 1000 5389 5365 20 0 22752 4208 - R+ pts/0 0:00 ps -lx
ubuntu@ubuntu:~/Desktop$ ps -lu ubuntu
F S UID PID PPID C PRI NI ADDR SZ WCHAN TTY TIME CMD
4 S 1000 2168 1 0 80 0 - 5434 - ? 00:00:01 systemd
5 S 1000 2169 2168 0 80 0 - 5365 - ? 00:00:00 (sd-pam)
4 S 1000 2182 2168 0 80 0 - 31050 ep_pol ? 00:00:00 pipewire
4 S 1000 2183 2168 0 80 0 - 26701 ep_pol ? 00:00:00 pipewire
4 S 1000 2185 2168 0 80 0 - 103977 do_pol ? 00:00:00 wireplumbe
4 S 1000 2186 2168 0 80 0 - 31136 ep_pol ? 00:00:00 pipewire-p
4 S 1000 2187 2168 0 80 0 - 81395 do_pol ? 00:00:00 gnome-keyr
4 S 1000 2192 2168 0 80 0 - 2673 ep_pol ? 00:00:00 dbus-daemo
4 S 1000 2211 2158 0 80 0 - 61202 do_pol tty2 00:00:00 gdm-x-sess
4 S 1000 2213 2211 1 80 0 - 78881 ep_pol tty2 00:00:14 Xorg
4 S 1000 2238 2211 0 80 0 - 76833 do_pol tty2 00:00:00 gnome-sess
4 S 1000 2320 2168 0 80 0 - 95748 do_pol ? 00:00:00 at-spi-bus
4 S 1000 2327 2320 0 80 0 - 2370 ep_pol ? 00:00:00 dbus-daemo
4 S 1000 2337 2168 0 80 0 - 40663 do_pol ? 00:00:00 gcr-ssh-ag
4 S 1000 2338 2168 0 80 0 - 25161 do_pol ? 00:00:00 gnome-sess
4 S 1000 2345 2168 0 80 0 - 80847 do_pol ? 00:00:00 gvfsd
4 S 1000 2355 2168 0 80 0 - 117204 futex_ ? 00:00:00 gvfsd-fuse
4 S 1000 2391 2168 0 80 0 - 169278 do_pol ? 00:00:00 gnome-sess
```

4.

```
0 R 1000 5391 5365 99 80 0 - 5688 - pts/0 00:00:00 ps
ubuntu@ubuntu:~/Desktop$ ps -lp 1
F S UID PID PPID C PRI NI ADDR SZ WCHAN TTY TIME CMD
4 S 0 1 0 0 80 0 - 5842 - ? 00:00:04 systemd
ubuntu@ubuntu:~/Desktop$ top -d 0.5

top - 09:19:24 up 25 min, 1 user, load average: 0.12, 0.12, 0.44
Tasks: 196 total, 1 running, 195 sleeping, 0 stopped, 0 zombie
%Cpu(s): 14.6 us, 2.1 sy, 0.0 ni, 83.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 1968.7 total, 205.2 free, 1154.2 used, 837.7 buff/cache
```

5.

```
20 1000 20 0 0 0 0 0 0 0 0 0 0.0 0.0 0.00.00 kaudito
ubuntu@ubuntu:~/Desktop$ top -u 999

top - 09:20:29 up 26 min, 1 user, load average: 0.04, 0.09, 0.41
Tasks: 196 total, 1 running, 195 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.8 us, 1.7 sy, 0.0 ni, 94.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 1968.7 total, 205.2 free, 1154.2 used, 837.7 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 814.5 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
  999
```

6.

```
bash: fg: %1: no such job
ubuntu@ubuntu:~/Desktop$ gedit &
[1] 5843
ubuntu@ubuntu:~/Desktop$ fg %1 gedit
bash: fg: job has terminated
[1]+  Done                  gedit
ubuntu@ubuntu:~/Desktop$
```

7.

```
ubuntu@ubuntu:~$ ps
  PID TTY          TIME CMD
 5973 pts/0        00:00:00 bash
 5981 pts/0        00:00:00 ps
ubuntu@ubuntu:~$ kill -STOP 5973
^C^Z^C^C^A^Z
```

8.

```
auto bash
ubuntu@ubuntu:~$ kill -CONT 5973
bash: kill: (5973) - No such process
ubuntu@ubuntu:~$ nice -n 10 gedit &
```

9.

```
ubuntu@ubuntu:~$ nice -n 10 gedit &
[1] 6105
ubuntu@ubuntu:~$ ps -l
F S  UID      PID   PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000     6083   6071  0  80   0 -   5025 do_wai pts/0        00:00:00 bash
4 S  1000     6105   6083  3  90  10 -  197607 do_pol pts/0        00:00:00 gedit
0 R  1000     6116   6083  99  80   0 -   5688 -          pts/0        00:00:00 ps
ubuntu@ubuntu:~$
```

10.

```
bash: kill: (5973) - No such process
ubuntu@ubuntu:~$ nice -n 10 gedit &
[1] 6105
ubuntu@ubuntu:~$ ps -l
F S  UID      PID   PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000     6083   6071  0  80   0 -   5025 do_wai pts/0        00:00:00 bash
4 S  1000     6105   6083  3  90  10 -  197607 do_pol pts/0        00:00:00 gedit
0 R  1000     6116   6083  99  80   0 -   5688 -          pts/0        00:00:00 ps
ubuntu@ubuntu:~$ sudo renice -n -15 -p 6150
renice: failed to get priority for 6150 (process ID): No such process
ubuntu@ubuntu:~$ sudo renice -n -15 -p 6105
6105 (process ID) old priority 10, new priority -15
ubuntu@ubuntu:~$ ps -l
F S  UID      PID   PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000     6083   6071  0  80   0 -   5025 do_wai pts/0        00:00:00 bash
4 S  1000     6105   6083  0  65 -15 -  197607 do_pol pts/0        00:00:00 gedit
0 R  1000     6132   6083  99  80   0 -   5688 -          pts/0        00:00:00 ps
ubuntu@ubuntu:~$
```